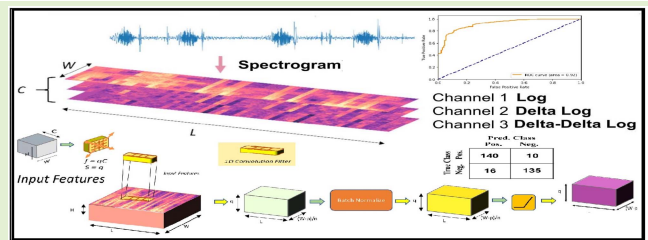


SpectroCardioNet: An Attention-Based Deep Learning Network Using Triple-Spectrograms of PCG Signal for Heart Valve Disease Detection

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Abstract—The phonocardiogram (PCG) signal is used for the early detection of cardiovascular diseases (CVDs) as it captures the heart sound characteristics. In this article, a spectral attention-based deep learning network is proposed for the automatic detection of cardiac disease from the spectrograms of PCG signals, namely SpectroCardioNet. From a given PCG audio signal, in view of simultaneously utilizing both time- and frequency-domain information, spectrograms, delta-spectrograms, and double-delta-spectrograms are generated. The extracted triple-spectrogram representation is applied in the proposed network as a three-channel 2-D input, where it passes spectral and sequential feature paths. In the spectral feature path, a spectral attention block (SAB) is designed to emphasize some regions in the spectrograms based on a deep attention network and its output is then processed through the spectral pattern detectors (SpPDs). On the other hand, in order to extract the temporal behavior of the frequency components of the spectrograms, a 1-D convolution-based sequential feature extractor is also proposed. Extensive experimentation is carried out on two standard PCG datasets and very satisfactory performance is achieved in comparison to that obtained by some existing methods.

Index Terms—Attention, cardiovascular disease (CVD), deep learning, neural network, phonocardiogram (PCG), spectrogram.



I. INTRODUCTION

PHONOCARDIOGRAM (PCG) signals are the sound waves corresponding to the mechanical activity of the heart, which represent the time-domain representation of the beat patterns of the heart. Because of its ease of acquisition and distinguishable variations in the case of different heart diseases, PCG is being widely used to detect cardiovascular diseases (CVDs). Moreover, by using the digital stethoscope, beating patterns can be easily visualized from the graphical representation of the PCG signals [1]. Since this is a sequential signal, many signal processing techniques have been used to extract features from the PCG signal as well as to segment the PCG signal, and various types of machine learning algorithms are used for the purpose of classification. Handcrafted features extracted through continuous wavelet transform (CWT), dis-

crete wavelet transform (DWT), S-transform, fast Fourier transform (FFT), and so on have been used and have been fed directly to different classifiers for classification purposes [2], [3]. In [4], spectral and analytical shape-based feature descriptors are used along with machine learning classifiers for PCG segment recognition. The K -means clustering, support vector machines (SVMs), random forest (RF), multiclass composite classifiers (MCCs), k -nearest neighbor (KNNs), and so on are some examples of the classifiers that have been reported in many articles. Depending on the various statistical/spatio-temporal nature of the features, the performance of the classifiers has been published [5], [6], [7]. In [8], the raw PCG signals are downsampled and passed through Savitzky–Golay filter to eliminate the high-frequency noises. The overall accuracy of 85.65% is claimed with the combined use of the 1-D convolutional neural network (CNN) and the feed-forward neural network (FNN) [8] for normal/abnormal class identification. Some empirical study on abnormal heartbeat detection of PCG signals is also tested on recurrent neural networks (RNNs) in the presence of noise [9]. In [10] and [11], segmentation problems are addressed with U-net and with a deep RNN (DRNN), respectively. Apart from this, long short-term memory (LSTM) networks, which are capable of

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dealing with sequence prediction problems, have been used in conjunction with genetic algorithms (GAs) to minimize prediction errors [12]. In [13], an end-to-end deep neural network (DNN) is proposed using a simple CNN model that offers moderate classification performance. The incorporation of DNNs can eliminate the need for handcrafted features from the system and introduce nonlinearity, which eventually improves the performance of the system. The CNN-based deep learning schemes are proved to be highly effective as a successful diagnostic system with an acceptable margin of accuracy.

The network proposed in [14] utilizes a CNN-LSTM architecture, which also uses a raw PCG signal input. On the other hand, instead of the 1-D PCG input, one may convert the input into 2-D image data and employ conventional CNN-based architectures. However, in this case, it not only requires transfer learning to initialize the weights of the network, but also involves an extremely large number of parameters [15]. Moreover, the amount of misprediction influences the need for a high-performing system that is independent of noise and evaluation of unnecessary parameters.

The existing approaches do not utilize both the spatial and temporal information of the PCG signal in the same network. Our proposed architecture is a study for comparative analysis between existing approaches and spectro-temporal features along with the attention mechanism. It shows an improved performance in sensitivity compared to other networks. This reveals the potential of incorporating both temporal and spatial features in designing biomedical diagnostic tools along with the attention mechanism to enhance the ability to correctly identify a patient with a disease. A brief description of the attention-based network will be available in Section II-B.

Since a PCG signal is analogous to speech signals, the application of speech recognition techniques could have good performance in PCGs as well. In this article, SpectroCardioNet, an end-to-end anomaly detection network is proposed that is powerful enough to distinguish abnormal heart sounds with a high level of precision. The proposed scheme overcomes the limitations of traditional approaches. Here, a triple spectro-temporal transformer (TSTT) for PCG signals is introduced that converts the 1-D representation of the PCG signal into a triple-channel spectrogram representation. In order to emphasize some regions in the triple-spectrogram input, first, a spectral attention block (SAB) is proposed followed by SpPDs. Moreover, a sequential feature extractor is designed to extract temporal features of various frequency components of the spectrogram. Extensive experimentation is carried out on publicly available datasets to investigate the comparative performance of the proposed network.

In this work, we present an end-to-end CVD identification tool through the fusion of three separate types of spectrogram operations: 1) a log-normal spectrogram; 2) a delta log-normal spectrogram; and 3) a double delta log-normal spectrogram. This work also focuses on the application of the attention scheme in the case of PCG signals, which has been widely

used in computer vision, speech signal processing, and natural language processing (NLP) tasks for the last couple of years. In particular, this work proposes a classification method that involves the following contributions.

- 1) Use of the triple spectrogram in DNN for classification.
- 2) Use of the pixel-based attention mechanism for fast model convergence and enhanced weights in the region of interest.
- 3) Use of spatial and temporal information in the same network.
- 4) Finally, projecting the memory requirement for the implementation of the model.

II. METHODOLOGY

In the proposed method, various spectro-temporal 2-D representations of the PCG signal are utilized in a deep learning network for precisely detecting cardiovascular abnormalities, and the network is termed as SpectroCardioNet. Major steps involved in the proposed scheme are shown in Fig. 1 with the help of a schematic. Traditional deep-learning-based approaches for the identification of abnormality in PCG signals include LSTM-based sequential networks that consider the correlation between successive frames of a given time-domain signal. In this case, only the time-domain information of a signal is considered ignoring the inherent useful frequency-domain information. On the contrary, the Fourier transform-based analysis of a signal mainly focuses on the frequency components of a signal without considering the between-frame time-domain correlation of the signal. Moreover, the frequency-domain representation of a signal does not provide time information of a specific frequency component.

In the sound wave of a PCG signal, during the “lub” or “dub” portion, voiced sound like time–frequency characteristics dominate (like the cvc, consonant-vowel-consonant structure) with the presence of some dominant frequencies. However, the other portions of the PCG signal (excluding the “lub” and “dub” regions) exhibit unvoiced sound like time–frequency characteristics without showing any dominant frequencies. For various types of cardiac problems, abnormalities can be found in time–frequency characteristics of the PCG signal. As a result, if the frequency characteristics of a PCG signal can be considered along with their time information, better identification of abnormality in PCGs can be performed.

One possible solution to obtain the time–frequency representation of a PCG signal is to use the spectrogram plots. In order to obtain a spectrogram representation, small overlapping frames of a time-domain signal are considered and a short-time Fourier transform (STFT) is performed in all these frames. This provides a frequency spectrum for each of the overlapping time frames. As a result, a spectrogram combines both time-domain and frequency-domain information in the same 2-D plot. Hence, in the proposed scheme, the spectrogram of the PCG signal is utilized. Moreover, in order to extract more information from a given spectrogram of the

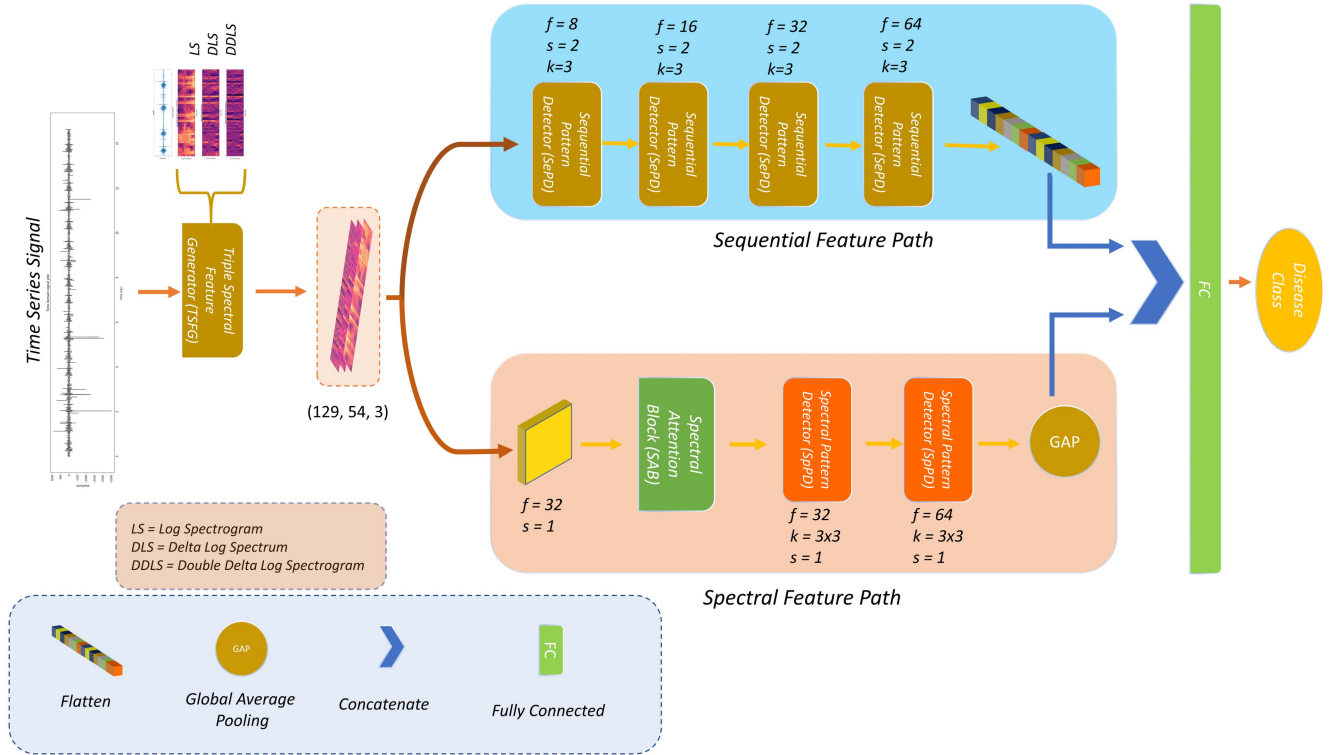


Fig. 1. Graphical overview of the complete scheme of the proposed SpectroCardioNet: TSTT transforms the 1-D time-domain representation into a triple-channel spectrogram. The two feature paths deal with both 1-D and 2-D patterns in the spectral representation, respectively. Here, f = filter size, s = strides, and k = kernel size in the corresponding convolution layers inside the blocks. The values of f , s , and k have been chosen in a trial-and-error fashion to achieve the best performance.

PCG signal with a view to differentiating various classes, two modified 2-D representations, derived from the spectrogram, are also introduced in the proposed scheme that will be discussed later.

Unlike conventional deep learning-based methods, instead of using only the raw PCG signal, in this research, 2-D spectrogram representations of the PCG signal are used as the input to the proposed CNN-based architecture for cardiac disease detection. In Fig. 1, it is shown that the proposed CNN model consists of two separate paths: 1) sequential feature path and 2) spectral feature path. In the sequential feature path, a sequential pattern block is designed using a 1-D CNN and implemented multiple times. In the spectral feature path, an attention network is first introduced followed by an SpPD, where a 2-D CNN is employed and the SpPD is implemented a couple of times. Finally, the features extracted from these two paths are concatenated and passed through a fully-connected (FC) layer to perform the classifications task. In what follows, each block of the proposed scheme is presented in detail.

A. Triple Spectro-Temporal Transformer

Let x be a time-domain digital signal with length N . Consider x having consecutive frames of length m , where $m \ll n$. Now, let X be a matrix where the previously mentioned frames are placed in the columns. Here, $X \in \mathbb{R}^{m \times (N-m+1)}$. $\hat{X}$ will be the spectrogram of x with a window size of m , where the columns of $\hat{X}$ are the discrete Fourier transform (DFT) of

the columns of X

$$\hat{X} = \bar{F}X$$

$$X = \frac{1}{m} \bar{F} \hat{X}.$$

Here, the columns of $\hat{X}$ are indexed in the time axis, the rows of $\hat{X}$ are indexed in the frequency axis of the spectrogram plot and F is a symmetric matrix used for linear transformation [16].

In order to obtain the spectro-temporal representation of a given PCG signal $x[n]$ of length N , in the proposed method, the spectrogram of $x[n]$ is considered. In the spectrogram operation, consecutive short-duration overlapping frames are constructed from $x[n]$, and DFT is used to obtain the frequency-domain representation of each frame. As a result, at some specific time instance, the spectrogram exhibits the magnitude of various frequency components. In Fig. 2, spectrogram representation of a sample PCG signal is shown.

In the case of our experimentation, we have chosen a frame length of 256 from the discrete time-series data that corresponds to a duration of 128 ms. The original sampling rate of the dataset was 8 kHz that was resampled to 2 kHz. The Nyquist rate after resampling is 1 kHz. Since all spectral features of the PCG signal are up to the range of 400–500 Hz, resampling at 2 kHz will not lose any important spectral feature. We chose an overlap of 32 ms among the consecutive frames while performing the spectrogram operation. The input-output sizes of the frames are shown in Fig. 2.

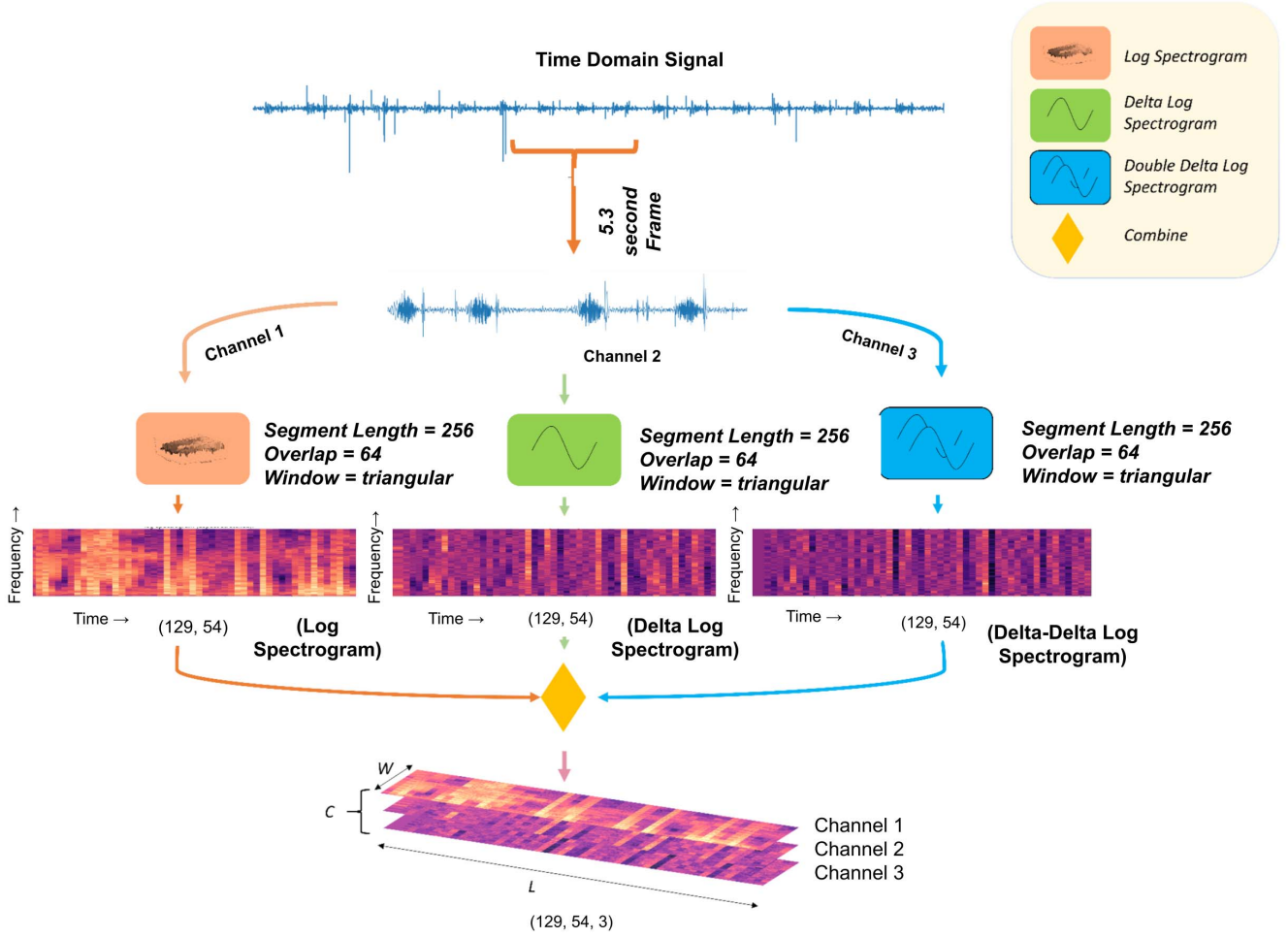


Fig. 2. Graphical representation of the TSTT module: A 5.3-s window is framed out from each audio signal and the later three channel paths include LS, DLS, and DDLS, each generating a 2-D representation with the shape of (129 and 54) of the 1-D time-domain signal. Finally, these three channels are merged together to form a representation that has a shape of (129, 54, and 3).

It is clearly observed from this figure that the 2-D spectrogram effectively depicts the time–frequency information of the PCG signal. Considering M number of frequency points and K number of sequential overlapping frames extracted from $x[n]$, the spectrogram of $x[n]$ can be represented as $\hat{X}[k]$, where each column of $\hat{X}[k]$ corresponds to the DFT of a frame of $x[n]$. In the proposed method, a 2-D spectrogram matrix is used for the purpose of feature extraction. Moreover, some alternative representations of the spectrogram are also explored to extract further information [17]. In this case, similar to the concept of a one-sample difference signal in 1-D time series, a delta spectrogram is introduced. The 2-D delta spectrogram matrix is constructed considering the difference between the actual spectrogram and the one-sample circular shifted version of the same spectrogram. The spectrogram is circularly shifted in the X -axis (time) that retains the size of the 2-D spectrogram but adds a slight shift of one unit in the spectrogram. Similarly, the double delta spectrogram is also introduced, which considers the difference between a delta-spectrogram and a sample circularly shifted delta-spectrogram. In the proposed method, a log-spectrogram (LS) is considered to accommodate a wide range of magnitude variations.

Accordingly, the delta LS (DLS) and double DLS (DDLS) are also considered

$$S_{\delta}(x, y) = S(x, y) - S(x - 1, y)$$

$$S_{\delta\delta}(x, y) = S(x, y) - S(x - 2, y)$$

where S_{δ} and $S_{\delta\delta}$ represent DLS and DDLS, respectively, and x and y represent columns and rows of the LS plot, respectively.

B. Spectral Feature Path

The LS that is going to be used for the purpose of feature extraction not necessarily contains the major frequency information of the PCG signal in all spatial regions. For example, according to the general characteristics of the PCG signal, major frequency contents exist in low-frequency regions, and those regions need to be paid more attention to during the feature extraction phase. Hence, we propose an SAB to emphasize some regions in the spectrograms based on a deep attention network. The output size of the SAB unit is the same as that of the input to the SAB unit. Afterward, the output of the SAB unit is passed through the proposed SpPD that is

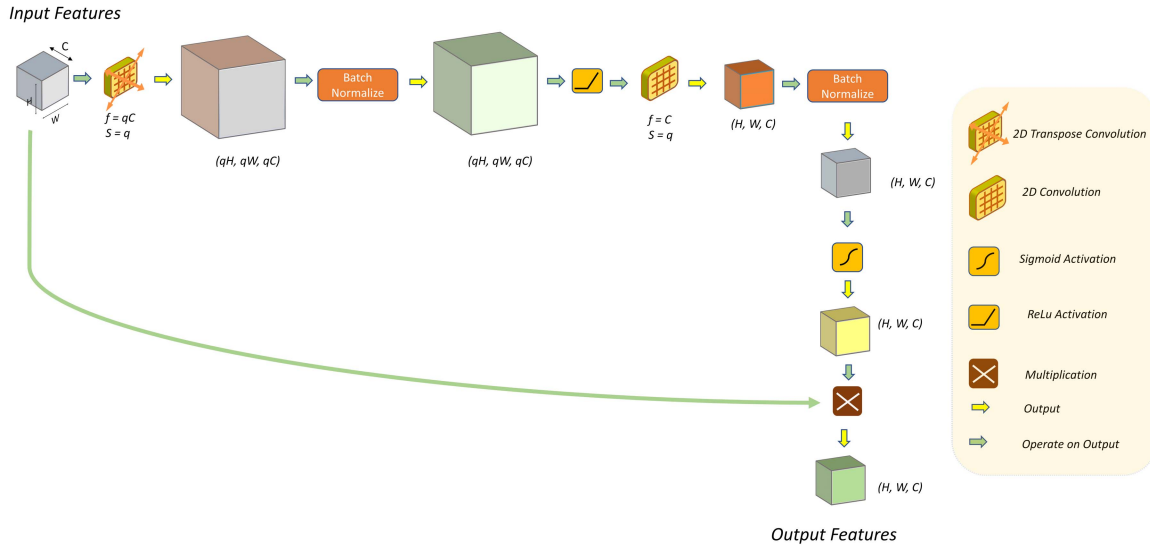


Fig. 3. Architecture of the SAB module.

designed to extract the features based on 2-D CNN operations. At the last stage of the spectral feature path, a global average pooling (GAP) layer is used to obtain a 1-D feature vector, which is later passed to the fusion segment of the overall network. In the following section, a detailed description of the proposed SAB unit and the proposed SpPD is presented.

1) Spectral Attention Block: The attention mechanism was first developed for the NLP in [18]. This mechanism was so successful that biomedical applications adopted this mechanism [19], [20]. It helps the network get faster convergence by putting more attention on the region of interest by a generalization of the predominant contextual features. It has been widely used in CNNs for image-related tasks as well as in the transformer-based architecture for NLP tasks. It has also shown great performance in speech modeling such as 1) automatic speech recognition and 2) text-to-speech models [21]. In this work, we present the performance evaluation of the attention architecture in heart disease detection along with CNNs. In the proposed scheme, an SAB is proposed that generalizes the region of interest of the spectrogram, as shown in Fig. 3.

The proposed attention mechanism operates on the input spectrogram with the shape of $(H, W, \text{and } C)$, where $H \times W$ is the spatial dimension and C is the number of channels. This module also takes an integer q as input that determines the level of expansion of the weights of the region of interest. At first, the input volume is passed through a transposed convolution layer with qC number of filters and strides of q . This amplifies the overall size of the input to $(qH, qW, \text{and } qC)$. This is the filtering mechanism for dimension expansion. To control the expansion and attain a faster convergence, the outputs are normalized through a batch normalization mechanism. At this point, the rectified linear unit (ReLU) activation function is employed to get linear transforms on the weights followed by a Conv2-D layer (with filters = C and strides = q). This restores the dimensions back to the size of $(H, W, \text{and } C)$. Finally, the

feature recalibration is performed through a sigmoid activation that normalizes the features in each dimension down to a probabilistic distribution of $R = [0, 1]$. This level creates a mask for the input feature space with high weights in the region of interest and low weights on the noncontextual information. Finally, the 2-D input to the SAB unit is then multiplied element-wise with the mask to emphasize the region of interest.

The reason behind using the ReLU activation function in this block as well as in other blocks frequently is that the ReLU activation function is faster. The derivative computation of the ReLU activation is also faster since, for negative inputs, the neurons are deactivated. From Fig. 1, it is evident that different blocks are repeated several times in two separate feature paths. So, faster computation of features and hence derivatives are highly important. During gradient descent, ReLU activation offers faster learning, since there are only two values available in the gradient of ReLU—0 and 1. This significantly eliminates the vanishing gradient problem found in other activation layers.

2) Spectral Pattern Detector: This module performs a number of operations on the SAB-operated spectrogram representations. It is expected that the SAB emphasizes some region of interest in the input spectrograms, and the 2-D convolution on the SAB-operated spectrograms will extract spatial correlation between the time and frequency features. The architecture of this module is portrayed graphically in Fig. 4.

This module takes a 3-D volume as an input and passes it through 2-D convolution filters with a kernel size of 3×3 and strides = 1. The number of filters is a variable parameter for the module. The output of the filter is as usual passed through a batch normalization layer, which ensures faster convergence of the weights and bias followed by the ReLU activation layer.

C. Sequential Feature Path

The sequential feature path includes four consecutive sequential pattern detector (SePD) modules that generate

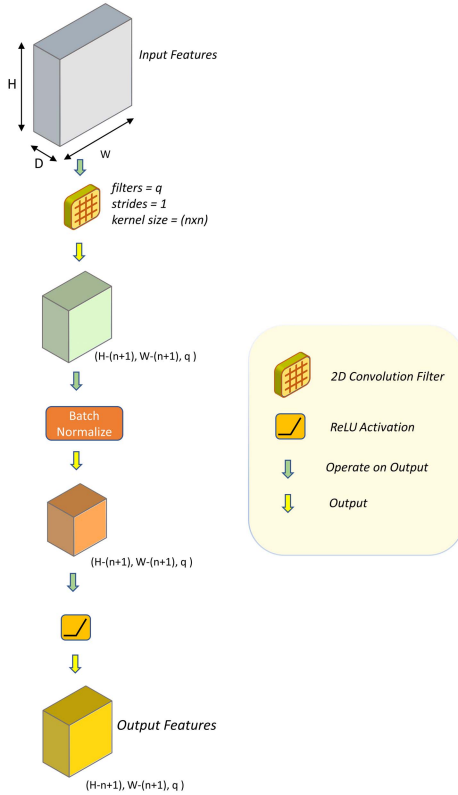


Fig. 4. Architecture of the SpPD module: the feature volume is passed through a Conv2-D layer followed by batch normalization and ReLU activation.

features from the input spectrogram through a 1-D convolution operation in the time axis of the spectrogram. At the final stage of this path, all the features are flattened to generate a 1-D feature vector that will be transferred to the fusion segment.

1) Sequential Pattern Detector: The SePD is almost similar to the SpPD module except for the type of convolution. In this module, the 1-D convolution operation is used instead of 2-D convolutions as used in the SpPD module. From an initial investigation of PCG signals, it is observed that the correlation between the frequency components of the spectrum has a high amount of contextual information which maps to the anomalies in the signals. Therefore, the Conv1-D operation is performed on the time axis of the outputs of TSTT. This module has been used multiple times in the sequential feature path as shown in Fig. 1. The complete representation of this module can be viewed in Fig. 5.

D. Fusion Segment

The 1-D feature vectors coming from the spatial and sequential feature paths are concatenated into a 1-D feature space. This array contains all the highly processed features and sends them to the final FC network of the system. The FC layer maps the features to the output labels and completes the classification task.

III. RESULTS AND DISCUSSION

In this section, the performance of the proposed scheme is analyzed on publicly available datasets to validate the effectiveness of the SpectroCardioNet. Results are also compared

with that obtained by some existing methods in the same datasets.

A. Dataset Description

The primary dataset that has been used in this validation process is the Physionet/CinC (2016) Challenge dataset publicly available at [22]. These PCGs were collected from different parts of the world both from the clinical and nonclinical environments. This database contains 3126 PCG records with a record duration ranging from 5.3 s to over 120 s. The recordings were sampled at a sampling rate of 2 kHz. These recordings were collected from different parts of the body: 1) aortic area; 2) pulmonary area; 3) tricuspid area; and 4) mitral area. The recordings are divided into two classes: 1) normal and 2) abnormal. The abnormal recordings were obtained from patients with the confirmation of CVDs. The normal PCG recordings were collected from healthy patients. But this dataset focuses only on the abnormality rather than the classification of heart diseases. To test the performance of the network in the classification of particular heart diseases, a secondary dataset was used that is publicly available in this GitHub repository [23]. This dataset contains 1000 PCG recordings in (.wav) format with five different classes: 1) normal (N); 2) aortic stenosis (AS); 3) mitral regurgitation (MR); 4) mitral stenosis (MS); and 5) mitral valve prolapse (MVP). The recordings were sampled at a sampling rate of 8 kHz. This dataset was resampled at 2 kHz since the frequency range of our interest lies in the range of 0–1 kHz.

B. Evaluation Metrics

For the purpose of performance measurement, the following evaluation metrics are used:

$$\begin{aligned} \text{Sensitivity} &= \frac{TP}{TP + FN} \\ \text{Specificity} &= \frac{TN}{TN + FP} \\ \text{F1 score} &= \frac{2 \times TP}{2 \times TP + FP + FN} \\ \text{Accuracy} &= \frac{TP + TN}{TP + FP + TN + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \end{aligned}$$

where TP, TN, FP, and FN represent the numbers of true positive, true negative, false positive, and false negative, respectively.

C. Experimental Setup

The Physionet dataset includes PCG recordings sampled at 2 kHz with varying duration ranging from 5.3 to 120 s. In the proposed scheme, keeping in mind the minimum recording duration, we consider the first 5.3-s window from each PCG data, which maintains consistency in the input data. Since the PCG recordings are sampled at a sampling rate of 2 kHz, the possible frequency components that can be analyzed through the spectrogram are up to 1 kHz. Since most of the dominant frequency components of PCG signals lie within the range of

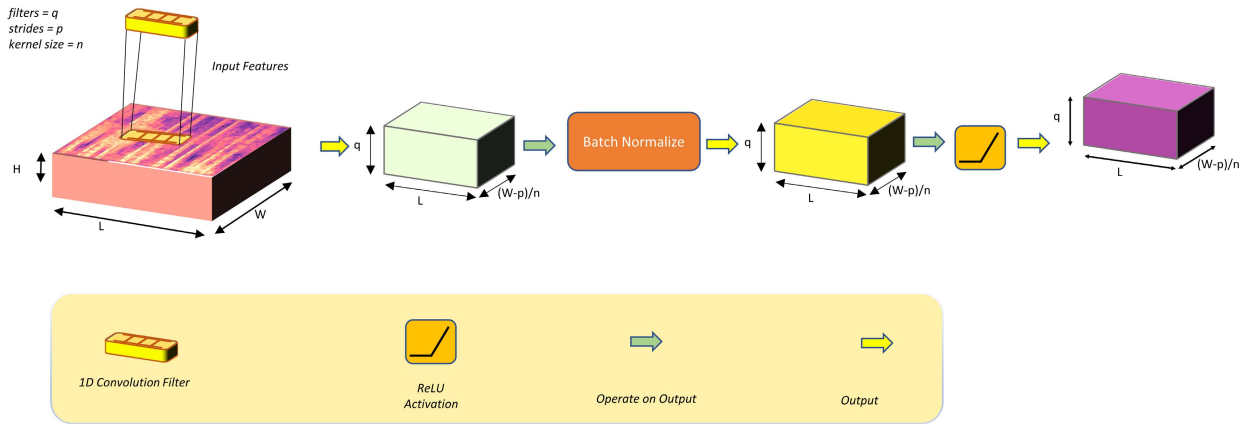


Fig. 5. Architecture of the SePD module: the feature volume is passed through a Conv1-D layer followed by batch normalization and ReLU activation.

20–200 Hz, the obtained frequency range is sufficient enough for the following tasks.

The Physionet/CinC 2016 Challenge dataset provides a separate set of 301 recordings that are chosen and marked as challenging recordings by the dataset provider. These recordings are also present in the provided training dataset. Hence, these 301 recordings are identified, removed from the training set, and the remaining 2825 PCG recordings are used in the training process. The 301 challenging recordings that were mentioned before are chosen as the test set. The training dataset (2825 PCG recordings) was divided randomly into separate training and validation sets at a ratio of 80% and 20%, respectively.

The experiment was performed on an NVidia K80 GPU with the tensorflow and Keras framework as a backend engine. For the Physionet/CinC 2016 Challenge dataset, the binary cross-entropy function is used for loss calculation, while the categorical cross-entropy function is used for the five-class classification problem on the secondary dataset [23]. In both cases, an adaptive learning rate optimizer was used with a learning rate of 0.008 followed by a learning rate reducer callback function to ensure the convergence of the network.

D. Performance Comparison

The comparative performance analysis of the proposed scheme with some existing methods on the Physionet/CinC 2016 Challenge dataset is presented in Table I. It can be clearly observed that the proposed method offers a very satisfactory performance in terms of all the measurement criteria, which justifies the effectiveness of the proposed method for the accurate detection of anomalies in the heart from PCG signals in medical applications. In [24], a 5-s window was chosen similar to the proposed scheme in this article. Wavelet decomposition, Hilbert transform, homomorphic filtering, and power spectral density (PSD) were used as features and the extracted features are provided to a KNN classifier. Our proposed network is not dependent on so many handcrafted features, which makes it handier.

In [31], feature extraction is performed through wavelet fractal transformation, and the features are classified through a twin SVM classifier. In [25], a novel semi-non-negative matrix factorization network (SNMFNet) was proposed that mainly focused on a dimensionality reduction method. Dimensionality

TABLE I

PERFORMANCE COMPARISON IN THE PHYSIONET/CINC 2016 CHALLENGES DATASET [22]

Methods	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 score (%)
SNMFNet [25]	-	89.18	77.66	-
AlexNet+SVM [26]	85.00	-	-	-
DNN [27]	85.65	86.73	85.74	-
DNN [28]	85.00	-	-	-
TFDDL [29]	85.16	-	-	-
Proposed method	91.36	93.33	89.40	91.50

TABLE II

PERFORMANCE OF THE PROPOSED NETWORK ON THE DATASET AVAILABLE IN GITHUB [23]

Methods	No. of Classes	Classifier	Accuracy (%)
WSST [5]	4	RF	95.13
MFCC [7]	5	SVM	91.60
DWT [7]	5	SVM	92.30
IFE [30]	5	RF	94.80
IFE [30]	5	KNN	91.60
Proposed method	5	DNN	97.77

		Pred. Class				
		AS	MR	MS	MVP	N
True Class	Pos.	43	1	0	0	0
	Neg.	0	44	0	0	0
	AS	0	0	35	2	0
	MR	0	1	0	30	0
	N	0	0	0	0	44

Fig. 6. Confusion matrix: the Physionet/Cinc-2016 dataset (left) and the Github dataset (right).

reduction along with expansion provides better performance, which can be observed in our proposed SAB unit.

For better visualization of the class-specific performance, the confusion matrix for both datasets is also presented in Fig. 6. For the Physionet dataset [22], it is evident that there are only 5.31% false positive cases and 3.32% false negative predictions in the test set. The receiver operating characteristic (ROC) for this dataset is also depicted in Fig 7. A comparative performance for different methods on this dataset is shown in Table II. The performance of the proposed method is also tested on a secondary dataset [23] that deals with multiclass CVDs. Apart from the accuracy, other

TABLE III
PERFORMANCE OF THE PROPOSED NETWORK ON DATASET [23]

Class	Accuracy	Precision	Recall	F1 Score
Normal (N)	99.85±1.4%	98.49±0.67%	98.22±0.98%	97.68±0.8%
Aortic stenosis (AS)	97.11±1.56%	96.12±0.23%	97.36±1.2%	98.44±1.6%
Mitral regurgitation (MR)	96.54±0.6%	96.33±0.65%	96.73±0.8%	95.37±0.88%
Mitral stenosis (MS)	97.34±1.2%	95.56±0.7%	95.23±0.98%	96.38±0.53%
Mitral valve prolapse (MVP)	98.01±0.59%	96.34±0.8%	95.77±0.67%	95.13±0.14%
Avg	97.77±1.07%	96.57±0.61%	96.66±0.93%	96.6±0.80%

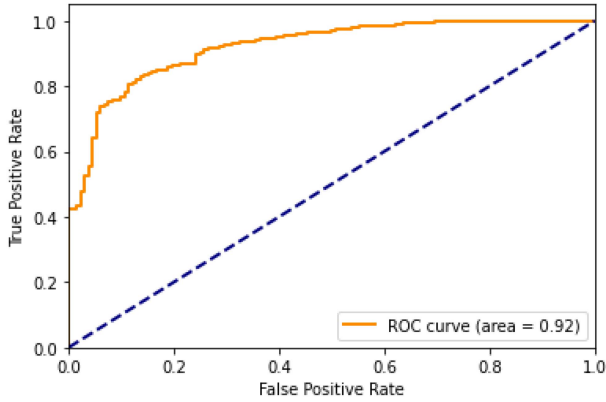


Fig. 7. ROC curve for the Physionet/CinC-2016 dataset.

performance measures obtained for the proposed methods are 1) precision; 2) recall; and 3) F1 scores, respectively, 96.57%, 96.66%, and 96.60%. In this dataset, satisfactory performance is also achieved by the proposed method. In this case, a fivefold cross-validation approach is adopted. The results can be seen in Table III.

In [29], a time–frequency domain-based approach along with the CNN has been used. It can be clearly observed that adding an attention scheme and three separate types of spectrogram mechanisms provided significant improvement in the performance. In conventional methods, a spectrogram of the PCG signal is considered as a color image consisting of three channels, and traditional deep learning networks for image processing are employed as a backbone with transfer learning. On the contrary, in the proposed scheme, the spectrogram is considered as a single-channel 2-D input, and a comparatively light new network is designed, which does not require transfer learning. The proposed SpectroCardioNet has 2.4 M training parameters resulting from a total of 25 layers altogether that is very low in comparison to the traditional image-handling networks, such as GG16 (138 M), ResNet50 (25.6 M), EfficientNetB5 (118 M), and DenseNet (121 M) [https://keras.io/api/applications/]. The proposed SePD and SpPD blocks are repetitive in the proposed architecture. Adding more blocks effectively increased the size of the network, but did not significantly improve its performance of the network.

IV. CONCLUSION

The proposed SpectroCardioNet utilizes triple spectrograms as the input and applies spectral attention-based modules for very fast and robust performance in cardiac disease classification. The performance of the proposed method is found very satisfactory even in the noisy data. Unlike the conventional methods, in the proposed method, the spectrogram is consid-

ered as the 2-D input, and no transfer learning is required for the proposed network. As a result, the number of trainable parameters is found very low in comparison to some traditional image-handling networks. Incorporation of the proposed spectral attention unit on the triple-spectrogram input followed by spectral pattern detection and use of the sequential feature path play a vital role in getting a satisfactory performance by the proposed network. Moreover, the use of delta and double delta spectrogram representations as the input on top of the regular spectrogram representation is also a contributing factor in getting better performance. According to our experiments, the estimated memory size for the model is 22 MB. This memory size can easily be provided through single-board computers such as Raspberry-Pi 4 or Beaglebone Black. This will allow the manufacturing of fully portable devices incorporating our model. Besides, this could also be done through a client–server approach in which the model will run on a remote server and the local data could be sent to the server through ESP8266- or ESP32-based microcontrollers. These low-power microcontrollers support the MQTT protocol—which is a client–server-based publish–subscribe messaging protocol. We could utilize such low-powered microcontrollers to send the locally captured PCG signals from the patient to the cloud where the disease could be diagnosed through our prediction model and the prediction results could be sent to the microcontrollers through the Internet. This can allow the manufacturing of the low-cost clinical diagnostic tool by utilizing our method. Because of its very satisfactory classification performance and low computational burden, the proposed network is expected to be a good candidate for real-life clinical applications. In this work, the TSTT was designed with the STFT method. But other methods such as chirplet transform are yet to be explored along with the current scheme instead of STFT.

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